Winter cycling on rue de Terrebonne and the days snow is loaded

What the borough's counter, Environment Canada and the City's snow-removal records show for winter 2025–26

Richard Hoge, Côte-des-Neiges–Notre-Dame-de-Grâce – October 13, 2026
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Summary

The borough of Côte-des-Neiges–Notre-Dame-de-Grâce has announced that one direction of the protected bike lane on rue de Terrebonne will be converted to parking each winter, December 1 to March 31. The engineering mandate behind the announcement also listed a narrower option, closing the lane "en période de déneigement seulement" [01, p. 10]. This note measures what each option would have meant for cyclists last winter, using three public records: the borough's own continuous bicycle counter at Terrebonne and Kensington, released under access request 2026-300; daily weather from Environment Canada; and the City of Montréal's open record of every truckload of snow hauled from the borough. Nothing is modelled; every figure is a count on a day that happened.

Two things stand out. Most winter days have no snow, those are the days with the most cycling, and on those days a parking lane in place of the bike lane serves no snow-related purpose while every eastbound trip is ridden in a traffic lane between parked cars. And on the days that do have snow, cycling continues at close to its usual winter level; on those days the lane would be closed under either option, and the trips are made on the roadway at its worst.

1. **Last winter 4,891 bicycle trips were counted at Kensington between December 1 and March 31, 2,783 of them eastbound.** Under an all-winter conversion every eastbound trip is a trip in mixed traffic. Under a closure limited to snow-loading days, the number is 410 on the borough's loading days (15%) and 548 on the loading days of the contract sector that contains Terrebonne (20%).
2. **Most winter cycling happens on days with no snow.** 1,699 eastbound trips, 61%, were on the 67 days with no snowfall at all; 83% on days with less than 2.5 cm.
3. **The result does not depend on the winter.** Winter 2025–26 was light: 7.3 million m^3 of snow hauled city-wide, the second-lowest of the eleven seasons in the City's record. Applying each of those eleven seasons' loading calendars to last winter's daily counts gives a storm-day share of 9% to 24%, median 18%.
4. **Within deep winter, loading days carry about the same cycling as other days.** From December to February eastbound trips averaged 17 a day on the borough's loading days and 18 on other days; the full-season difference (17 against 25) comes from March, when use rises and snow is scarce. Winter cyclists ride through loading operations; they do not stay home for them.
5. **The counter's westbound channel degraded around March 1, 2026.** A camera on the same street confirms it. The figures above treat the eastbound counts as lower bounds from that date; on the firm window, December 1 to February 28, the storm-day shares are 23% and 28% and the zero-snow share 48%. The conclusions are the same on either window.

1 Data

Bicycle counts. The borough operates an Eco-Counter ZELT EVO inductive-loop counter on Terrebonne at Kensington, in service since July 26, 2025. Its hourly record by direction, July 26, 2025 to May 11, 2026, was released on June 4, 2026 under access request 2026-300 (raw/); the equipment documentation

was released under request 2025-697. Summed to days, the record matches exactly the daily totals the borough released on October 9, 2026 in file 2026-581 [01, p. 74–81]: December 1 to March 31, 4,891 trips, 2,783 eastbound and 2,108 westbound. Over the counter's first full year, July 26, 2025 to July 25, 2026, the same daily table gives 135,276 bicycle passages (371 a day) and 501,817 motor vehicles (1,375 a day): one bicycle for every 3.7 motor vehicles, or 21% of all vehicles counted. The bicycle figure is a lower bound: May 29–31, 2026 are three days of zero (an outage), and the westbound channel's detection degraded from about March 1, 2026 (see Limits). Eastbound is taken here as the direction that would lose its lane: the councillor for Loyola said, in answer to questions at the October 6 meeting and earlier to the APC-NDG, that the lane running in the same direction as motor traffic would be converted, which at Kensington is eastbound (west of Cavendish, where the one-way reverses, it is westbound). If the westbound lane were converted instead, every figure scales by 2,108/2,783 and no conclusion changes.

Weather. Environment and Climate Change Canada daily data for Montréal/Pierre Elliott Trudeau International Airport (climate ID 7025251): total snowfall, snow on the ground, temperature.[1] One missing day (March 8) is treated as no snow.

Snow loading. The City publishes every snow-removal truckload with its loading time, contract sector, borough, dump site and volume (*Contrats et transactions de déneigement*, CC-BY 4.0).[2] Filtered to borough code CDN and summed by day, it gives the borough's loading days for every season since 2015–16. The City's sector polygons for 2025–26[3] place the lane in contract sector CDN-301 from its west end, just west of Belmore, to between Royal and Draper, about 1.8 of its 2.4 km, the counter included; the last blocks, Draper to Girouard (about 0.55 km), are in sector CDN-111, and the Girouard intersection lies on the boundary with CDN-112 (intersections from the City's géobase; the check is `terrebonne_sector.py`). A day with 100 or more loads in the borough, or 40 or more in the sector, is counted as a loading day; full operations run at several hundred loads a day and the tails below those thresholds are mop-up. In 2025–26 this gives 24 borough loading days in six episodes (December 12–16 and 28–31, January 8–9, 17–21 and 27–31, February 27 to March 2), matching the six operations the City announced, and 30 days for sector CDN-301, whose operations run a day or two longer.

2 Method

Let n_d be the eastbound trips counted on day d of the season S (December 1 to March 31, 121 days). Under an all-winter conversion the exposure is $E_{\text{all}} = \sum_{d \in S} n_d = 2{,}783$. Under a closure limited to a set of storm days $\mathcal{D}$, it is $E_{\mathcal{D}} = \sum_{d \in \mathcal{D}} n_d$. Since the City has not said how a storm closure would be defined, $\mathcal{D}$ is computed under several definitions, from a single snow day to every day with any loading activity in the borough. To test the dependence on the winter, each of the eleven seasons' CDN loading calendars is also applied, by calendar date, to the 2025–26 counts: the same cyclists, a different storm calendar.

3 Results

The storm days are a minority of the trips. On the borough's measured loading days a storm-only closure would have put 410 eastbound trips into mixed traffic, 15% of the 2,783 an all-winter conversion affects; on the sector's, 548 (20%); extended back to the triggering snowfall, 592 (21%); on every day with any loading activity at all, 1,129 (41%). Under the most generous snow-day rule in the table, 44%.

[1] ECCC historical climate data, https://climate.weather.gc.ca; the two annual files are in derived/.

[2] https://donnees.montreal.ca/dataset/contrats-transaction-deneigement

[3] *Secteurs de déneigement*, https://donnees.montreal.ca/dataset/secteur-deneigement.

Closure rule	Days	Eastbound trips	Share of all-winter
All winter, December 1 – March 31	**121**	**2,783**	**100%**
Borough loading days (≥100 loads)	24	410	15%
Sector CDN-301 loading days (≥40 loads)	30	548	20%
From triggering snowfall (≥2.5 cm) through borough loading	36	592	21%
From triggering snowfall through sector CDN-301 loading	44	770	28%
Any day with loading activity in the borough	59	1,129	41%
Any day with loading activity in sector CDN-301	36	600	22%
Any day with ≥2.5 cm of snow	24	462	17%
≥2.5 cm day and the day after	44	843	30%
≥5 cm day and four days after	66	1,219	44%
Any day with measurable snow	54	1,084	39%
Firm window, December 1 – February 28 (90 days; 1,626 eastbound trips)			
Borough loading days (≥100 loads)	22	380	23%
Sector CDN-301 loading days (≥40 loads)	26	450	28%
From triggering snowfall through sector CDN-301 loading	40	672	41%
Any day with ≥2.5 cm of snow	19	322	20%

Table 1: Eastbound bicycle trips at Terrebonne/Kensington that would have been ridden in mixed traffic under each closure rule, winter 2025–26. Counter: request 2026-300. Snowfall: ECCC, Montréal–Trudeau. Loading: City transactions dataset, borough CDN and sector CDN-301.

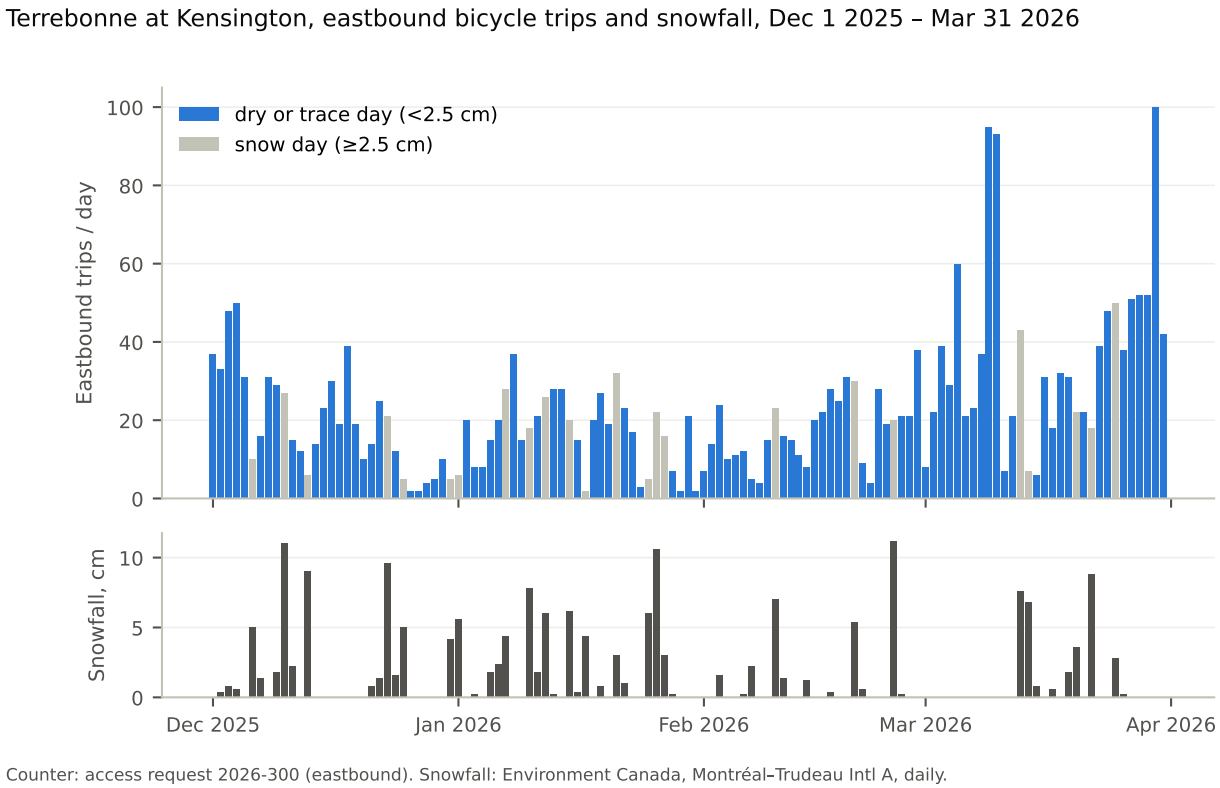


Figure 1: Daily eastbound trips (top) and snowfall (bottom), December 1, 2025 to March 31, 2026. Grey bars mark days with 2.5 cm or more of snow.

Season whose CDN loading calendar is applied	Loading days	Eastbound trips	Share
2015–16	21	481	17%
2016–17	30	550	20%
2017–18 (heaviest in the record, 18.0 M m^3)	42	639	23%
2018–19	40	673	24%
2019–20	24	443	16%
2020–21	21	399	14%
2021–22	27	527	19%
2022–23	27	517	19%
2023–24 (lightest, 6.5 M m^3)	12	247	9%
2024–25	27	503	18%
2025–26 (7.3 M m^3)	24	410	15%

Table 2: The 2025–26 daily counts under each season's borough loading calendar ($\geq$100 loads a day, aligned by calendar date). Only the storm calendar changes between rows.

Every definition leaves more than half of the all-winter exposure as trips on days when no storm closure would have applied.

Most winter cycling happens on dry days. Of the 2,783 eastbound trips, 1,699 (61%) were on the 67 days with no snowfall at all, and 2,321 (83%) on the 97 days with less than 2.5 cm. On the December–February window, 775 of 1,626 (48%) were on zero-snow days.

Cyclists ride on snow days and through loading operations. Over the season, mean eastbound trips were 25.4 on dry days, 20.7 on trace days and 19.2 on days of 2.5 cm or more, and 17.1 on the borough's loading days against 24.5 on other days. Most of that gap is March: on the December–February window the figures are 17.2 on dry days, 16.9 on $\geq$2.5 cm days, and 17.3 on loading days against 18.3 on other days. Day-to-day, trips correlate with mean temperature ($+0.54$ over the season) and with snow on the ground (-0.44) more than with the day's snowfall (-0.12): cold and accumulated snow suppress cycling more than the day's fall does. The exceptions are the heaviest storms: the January 27–31 operation, which followed 16.6 cm in three days, saw 2, 7 and 2 trips on three of its five days.

The winter does not change the answer. Winter 2025–26 was light, 185 cm at Trudeau against a long-run mean near 210, with no day above 11.2 cm and 7.3 million m^3 hauled city-wide, the second-lowest of eleven seasons (median 12.0; 2017–18, 18.0). The second table applies the other ten seasons' loading calendars to the same daily counts: the share ranges from 9% to 24%, median 18%, and the heaviest season on record gives 23%. At sector level the eleven seasons give 8% to 23%.

4 In short

On the dry days, the conversion costs cyclists and relieves no snow constraint. Sixty-one percent of last winter's eastbound trips (1,699; 48% on the December–February window) were ridden on days with no snowfall, when the street's parking supply is not reduced by snowbanks and no loading operation is under way. Under an all-winter conversion each of those trips is made in the single remaining traffic lane, between parked cars on both sides, with the hazards that configuration carries: less room to pass, and the opening of car doors along the whole block. Under a storm-only closure none of them is affected. What the conversion provides on those days is parking on a day when parking is not short of space for snow; whether that is worth the exposure is a judgement this note does not make, but the quantities on each side are now known.

On the snow days, cycling continues, and the exposure per trip is greater. Within deep winter, loading days carried as many eastbound trips as other days (17 against 18 a day, December to February). These are the days of reduced traction and visibility, when the margin between a bicycle and a passing vehicle matters most. Under either option the lane is unavailable on these days; the difference between the options is entirely in the dry days above. A closure confined to the loading days would thus place the trips in mixed traffic on exactly the days the roadway is worst, and nowhere else; the all-winter conversion adds the rest of the season to that.

Put as a hypothetical. If the current proposal had been in force last winter, and assuming that the new configuration would not have deterred cyclists from riding on Terrebonne, as was predicted at the October 6 council meeting, the following impacts could be posited:

1. on the 67 snow-less days, a total of 1,699 bike trips would have been shifted from a protected bike lane to a driving lane, possibly narrower than today's, with parked cars on both sides;
2. on the remaining 54 days, 1,084 bike trips made on days with snowfall would have been shifted to the shared driving lane.

The measure is confined to winter, yet over half of last winter's days, 67 of 121, had no snowfall. A dry day can still follow a storm with the snowbanks not yet cleared; counting every day from a triggering snowfall of 2.5 cm or more to the end of the loading operation that followed in Terrebonne's own clearing sector, CDN-301, as a day on which snow constrained parking gives 44 such days and 770 trips, and leaves 77 days (64%) and 2,013 trips (72%) on which the lane would have been parking with no snow on the street to justify it. The same rule on the borough's loading days, which end a day or two sooner, gives 36 days inside and 85 days (70%) and 2,191 trips (79%) outside (Table 1). The assumptions are those of the note: the trips are eastbound passages at Kensington, the direction the councillor said would be converted, and a floor on use of the street; snowfall is the day's total at Trudeau airport, and the snow-less days include days with snow on the ground; the driving lane's width under the winter layout is not in the released file.

5 Limits

The counter from March 1, 2026. The counter's eastbound and westbound totals are balanced to within a few percent in every month from July 2025 to February 2026 and diverge from March (70% eastbound in March, 59–61% in April and May). The citizen Telraam camera on Terrebonne at King-Edward, which observes the same roadway, agrees with the loop's eastbound channel through March but records westbound days of 80 to 190 bicycles where the loop records 10 to 40, after four months in which the loop counted 1.6 to 2 westbound bicycles for each one the camera saw. The loop's westbound detection therefore degraded around March 1, and its counts from that date are lower bounds. The March eastbound figures are consistent with the camera but are not vouched for separately, which is why the table also gives the December–February window. Asked on October 13, 2026 whether the counter was serviced, moved or re-tuned, the borough's engineering division replied on October 15 that nothing was done to the loop between mid-February and mid-March 2026, that the only recorded intervention is a battery replacement in early June with a brief interruption whose data were reconstituted by the manufacturer, that it sees no evidence of a technical fault, and that a snow or ice bank on the westbound side, the side next to the parked cars, may have altered where people rode. The weekly record places the break in the week of March 2 and shows the imbalance persisting at 58–61% eastbound through April and early May, after the snow was gone, while the camera's directional split over the same months barely changes; the westbound counts from March 1 therefore remain lower bounds here.

One winter, one station, one counter. The counter stands at one point on a 2.45 km lane; use varies along it. Trudeau airport is 12 km west of Terrebonne; day-level snowfall is well correlated across the island, but amounts on NDG streets can differ. The loading days are the borough's and sector CDN-301's; the lane's last blocks, Draper to Girouard, lie in sector CDN-111 and are outside the sector figures, and sector boundaries may have changed between contracts (only the 2025–26 polygons were checked; CDN-301 appears as 301A and 301B in 2019–20 and is combined). The multi-season table holds the cyclists fixed and changes only the calendar; in a heavier winter fewer people would have ridden on the storm days themselves, which makes those rows conservative in the other direction.

What this note does not measure. The parking side: how many spaces are lost to snowbanks on which days, and how many vehicles need them. The City's record contains no such count. This note measures what the cyclists' side of an all-winter and a storm-only closure would have been; the comparison of benefits is for the reader.

6 Files

All inputs and the script that produces every number above are in 2026-300/derived/: the counter record summed to days by direction; the merged daily table with weather and loading activity; the City's loading record summed by season, day, borough and sector; the two ECCC annual files; the sector lookup; and `exposure_analysis.py`, which regenerates the tables and the figure from those files. The counter's hourly workbook and the borough's response letter are in 2026-300/raw/, exactly as received.